# Per-Client Adaptive Differential Privacy for Federated Learning in Heterogeneous UAV Swarms

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Abstract—Federated learning (FL) with differentially private stochastic gradient descent (DP-SGD) is the standard approach for privacy-preserving model training across a fleet of unmanned aerial vehicles (UAVs), but UAV cohorts are small and their on-board data holdings are highly heterogeneous, a regime standard DP-FL calibration handles poorly. We show that calibrating a single, shared noise multiplier $\sigma$ to a fleet's smallest client, the standard practice, forces every larger client to train under far more noise than its own formal privacy guarantee requires, and that this is not a minor inefficiency: on CIFAR-10 at a genuinely valid per-client $\varepsilon \leq 10$ guarantee, uniform-$\sigma$ DP-FL in a 20-UAV cohort achieves only 17.3% ± 3.9 mean final accuracy across 5 seeds, frequently indistinguishable from chance. We propose per-client adaptive noise calibration: since each client's privacy loss depends only on its own data, each client is assigned the smallest $\sigma$ its own sample count permits, holding an identical formal per-client guarantee. This raises CIFAR-10 accuracy to 30.7%±6.6 (+13.4 percentage points, 5 seeds) and EuroSAT accuracy to 54.5% ± 4.8 (+16.3pp, 3 seeds, winning every seed tested), while providing no benefit on a severely class-imbalanced real UAV disaster-imagery dataset (AIDER), a result that itself rules out noise magnitude as that dataset's bottleneck. We integrate this mechanism into a full UAV data-sharing stack: a Proof-of-Authority blockchain with Byzantine/Sybil/free-rider defences, a Stackelberg-game incentive layer, and a multi-agent reinforcement learning (MARL) layer in which each UAV learns its own participation policy via federated Q-learning. Reaching a baseline against which the proposed mechanism could be honestly evaluated required finding and fixing three layered privacy-accounting errors, including one that invalidated a previously-reported headline result; we report this history explicitly as a reproducibility contribution in its own right.

Index Terms—Federated learning, differential privacy, UAV networks, privacy accounting, multi-agent reinforcement learning, blockchain, incentive mechanisms.

## I. Introduction

Unmanned aerial vehicle (UAV) swarms increasingly act as mobile sensing and computing platforms for time-critical applications such as emergency response, where a fleet collaboratively trains a shared model over locally sensed data (imagery, telemetry) without transmitting raw observations. Federated learning (FL) [1] is the standard mechanism for this collaborative training, and differential privacy (DP), typically via DP-SGD [2] with a Rényi DP (RDP) accountant [3], [4], is the standard mechanism for bounding what an adversary can infer about any single UAV's underlying data from its shared gradient updates.

A UAV fleet differs from the data-centre FL settings DP-FL was designed for in two respects that standard practice does not accommodate well. First, cohorts are small, tens of UAVs rather than hundreds or thousands of mobile phones, so each client's gradient update carries proportionally more signal (and more risk) than in a large-population setting. Second, UAV data holdings are highly heterogeneous: flight duration, sensor coverage, and mission assignment all vary, so one UAV's on-board dataset may be an order of magnitude larger than another's. Standard DP-FL practice calibrates a single noise multiplier $\sigma$ to the population's smallest client, the one requiring the most noise for a given formal guarantee, and applies that same $\sigma$ to every client. In a small, heterogeneous UAV cohort, this forces every larger client, who could tolerate substantially less noise under an identical formal guarantee, into noise calibrated for a client that may hold an order of magnitude less data.

This paper makes three contributions:

1) We show, across two datasets and eight independent random seeds, that this uniform-$\sigma$ practice is not merely suboptimal but close to disabling learning entirely at a genuinely valid per-client privacy guarantee in a representative small-cohort UAV setting, and we introduce per-client adaptive noise calibration, a mechanism that exploits the already-independent nature of per-client DP accounting to give each client only the noise its own data size requires, recovering substantial learning at an identical formal guarantee (Section IV, VI).
2) We integrate this mechanism into a complete UAV data-sharing stack combining DP-FL, a Stackelberg-game incentive layer with cryptographic proof of resource sharing, a Proof-of-Authority blockchain with Byzantine/Sybil/free-rider defences, and a multi-agent reinforcement learning (MARL) layer in which each UAV learns its own participation policy via federated Q-learning (Section III).
3) We report, as a reproducibility contribution independent of the above, three layered privacy-accounting errors found and fixed over the course of this work, the last of which silently invalidated a previously-

reported headline result by deriving the population's worst-case client against a partition the actual simulation pipeline does not produce (Section V). Each error produced a plausible, internally consistent, and wrong conclusion; we report the full correction history because this failure mode is easy to repeat, and was repeated more than once in this project.

The remainder of this paper is organised as follows. Section II reviews related work in DP-FL, FL client selection, and UAV-based FL systems. Section III describes the system model and the full data-sharing stack. Section IV details DP-SGD privacy accounting and derives the adaptive calibration mechanism. Section V documents the three accounting errors this work found. Section VI presents the multi-seed experimental results. Section VII discusses limitations, and Section VIII concludes.

## II. Related Work

Federated learning and DP-SGD. FedAvg [1] established communication-efficient FL; DP-SGD [2] adds per-sample gradient clipping and Gaussian noise to bound each client's contribution, with the Moments/RDP accountant [3] and its subsampled extension [4] giving the tightest known composition bound for the subsampled Gaussian mechanism. Recent DP-FL work targets the accuracy cost of DP-SGD directly: DP-FedSAM [5] uses sharpness-aware minimisation to flatten the loss landscape against DP noise, and DP-FedPGN [6] uses a perturbed gradient norm for a similar effect; both report results on pools of hundreds of clients. Hybrid local/global DP schemes [7] relax the threat model to retain more accuracy. None of this work addresses noise allocation across a heterogeneous client population under a shared formal guarantee, the gap this paper targets: all calibrate one $\sigma$ for the whole population, as we show standard practice does, and improve the model or optimiser instead.

Client selection and weighting in FL. A substantial literature addresses which clients to select or how to weight their contributions under heterogeneity, typically for statistical heterogeneity (non-IID labels) or system heterogeneity (compute, bandwidth), rather than for reallocating a shared privacy budget. To our knowledge, exploiting the per-client independence of DP accounting to assign heterogeneous clients heterogeneous noise levels under an unchanged formal guarantee is not addressed in this line of work.

Byzantine robustness and Sybil defences. Coordinate-wise median and trimmed-mean aggregation [8], [9] and cosine-similarity-based Sybil detection [10] are standard defences against adversarial or duplicated clients; we adopt both as the security layer of the integrated stack (Section III).

UAV networks, FL, and MARL. UAV-assisted wireless networking [11] and UAV-based FL are both active areas, typically addressing channel modelling, trajectory optimisation, or energy constraints in isolation from formal privacy guarantees. Deep Q-networks [12] and federated reinforcement learning are likewise each individually established; combining per-agent DQN participation policies with federated Q-network sharing specifically under a DP-FL constraint, as in Section III's MARL layer, is comparatively underexplored.

## III. System Model

The system comprises $N$ UAVs performing 3D random-waypoint mobility [13] over a bounded area, communicating with a ground parameter server over a Rician air-to-ground channel. A UAV participates in a federated round only if its instantaneous channel quality exceeds a minimum SNR threshold $\gamma_{\min}$; this channel-aware eligibility gate, combined with a fixed participation fraction, typically admits 11 of 20 UAVs per round in our evaluation setting.

### A. Federated Learning with DP-SGD

Each participating UAV $i$ holds a local dataset of $n_i$ samples and trains locally for $E$ epochs with batch size $b$, using per-sample gradient clipping at norm $C$ and additive Gaussian noise scaled by its assigned noise multiplier $\sigma_i$ (Section IV). The parameter server aggregates via FedAvg, weighted by $n_i$.

### B. Incentive Mechanism

A Stackelberg game models the parameter server as a leader setting a reward rate $\lambda^*$, with UAVs as followers choosing effort (additional local epochs) to maximise their own net utility given their individual cost coefficient $c_i$. The closed-form social-welfare-optimal rate is $\lambda^* = (-1 + \sqrt{1 + 2\alpha H})/(2H)$ where $H = \sum_i 1/c_i$ and $\alpha$ is the server's quality valuation; each UAV's best response is $e_i^* = \lambda^*/c_i$. A Proof-of-Resource-Sharing (PoRS) scheme requires each UAV to commit a cryptographic hash of its gradient update, a Merkle root over its training batches, and measured compute time and bytes transmitted, verified on-chain before token rewards are distributed.

### C. Blockchain Security Layer

A lightweight Proof-of-Authority (PoA) consensus among a reputation-ranked validator subset avoids proof-of-work's energy cost, appropriate for battery-constrained UAVs. On-chain smart-contract logic maintains token balances and reputation, and three attack models are defended against: Sybil attacks (cosine-similarity clustering and a stake requirement, following [10]), Byzantine attacks (coordinate-wise median or trimmed-mean aggregation [8], [9], with a Z-score norm-outlier and cosine-deviation detector), and free-riding (gradient-norm thresholding with inter-round staleness checks).

### D. MARL Agentic Participation

Static participation scheduling (a fixed fraction of eligible UAVs per round) ignores per-UAV state: battery level, channel quality, local data quality, and accumulated reward history. Each UAV instead runs an independent deep Q-network (DQN) [12] that learns a participation policy over this state, with Q-network weights periodically federated across UAVs (a federated Q-learning step analogous to FedAvg applied to the policy rather than the task model), so experience propagates across the swarm without centralising state.

## IV. Per-Client Adaptive Differential Privacy

### A. The Uniform-$\sigma$ Pitfall

Each client's $(\varepsilon, \delta)$-DP guarantee is tracked by an independent RDP accountant, composing the per-step cost of the subsampled Gaussian mechanism over that client's own sequence of noisy gradient releases. A client's privacy loss depends only on what is revealed about its own data; there is no formal requirement that every client in a population share one noise multiplier. Standard practice nonetheless derives a single global $\sigma$ by finding the smallest value that keeps the population's worst-case (smallest) client within budget, assuming that client participates in every round, and applies it uniformly. For a client with $n_i \gg n_{\min}$ samples, the DP-SGD subsampling rate $q_i = b/n_i$ is correspondingly smaller, so that client's actual per-step privacy cost at the shared $\sigma$ is far below its own target, noise it did not need to spend to clear its own guarantee.

### B. Per-Client Adaptive Calibration

We instead assign each client $i$ its own noise multiplier $\sigma_i$, the smallest value such that $i$'s own RDP accountant, composed over its own sampling rate $q_i$ and step count for the full round budget $R$, stays within the shared target $(\varepsilon_{\max}, \delta)$ even under worst-case (every-round) participation:

$$\sigma_i = \min\{\sigma : \text{RDP-to-DP}\big(\textstyle\sum_{r=1}^{R} \text{RDP}_q(\sigma; q_i), \delta\big) \le \varepsilon_{\max}\}, \tag{1}$$

found by binary search against the real accountant (Algorithm 1). Each client trains locally with its own $\sigma_i$; FedAvg's existing sample-weighted aggregation then naturally emphasises the larger, now cleaner-signal clients, while the smallest clients, still necessarily the noisiest, retain their existing (small) aggregation weight. No change to the aggregation rule, the accountant's composition rule, or the formal per-client guarantee is required; only the noise each client is permitted to add changes.

## V. Three Privacy-Accounting Errors

Reaching a baseline against which adaptive calibration could be honestly evaluated required finding and correcting three layered accounting errors, each of which had, at the time, produced a confident and internally consistent but incorrect conclusion. We summarise them here as a methodological contribution independent of the mechanism in Section IV; full diagnostic detail is retained in the accompanying technical report.

**Algorithm 1** Per-client $\sigma$ calibration

**Require:** client sample count $n_i$, batch size $b$, local epochs $E$, round budget $R$, target $(\varepsilon_{\max}, \delta)$

$q_i \leftarrow b/n_i$; steps $\leftarrow E \cdot \lceil n_i/b \rceil \cdot R$
$(\sigma_{\text{lo}}, \sigma_{\text{hi}}) \leftarrow (0.05, 50.0)$
**while** $\sigma_{\text{hi}} - \sigma_{\text{lo}} > 10^{-3}$ **do**
  $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$
  $\varepsilon \leftarrow \text{RDPAccountant}(\sigma_{\text{mid}}, q_i, \text{steps}).\text{get_epsilon}(\delta)$
  **if** $\varepsilon > \varepsilon_{\max}$ **then** $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$
  **else** $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$
  **end if**
**end while**
**return** $\sigma_{\text{hi}}$

Error 1: step undercounting. An early implementation credited the accountant with the number of local epochs as the DP-SGD step count, rather than the true number of noisy minibatch releases ($E \times \lceil n_i/b \rceil$ per round), undercounting composition by roughly two orders of magnitude for a typical client.

Error 2: pooled accounting. A single shared accountant received every client's cost, so the reported $\varepsilon$ grew with population size rather than reflecting any individual client's actual privacy loss, a 100-client run's reported $\varepsilon$ tracked a 20-client run's by almost exactly the $5\times$ participant-count ratio. Corrected by giving each client an independent accountant and reporting the population guarantee as the maximum over clients, not a sum.

Error 3: wrong worst-case client. The $\sigma$ used to derive a previously-reported CIFAR-10 result was computed by a script that partitioned the dataset directly, omitting the random draws the actual simulation pipeline consumes (UAV network, mobility, and channel initialisation) before partitioning. This produced a different, more lenient worst-case client than the one the real pipeline actually instantiates; at the previously-used $\sigma$, the true worst-case client's real $\varepsilon$ over the evaluation's round budget was approximately 16, not the claimed $\le 10$. All CIFAR-10 results in this paper use the corrected, pipeline-consistent calibration.

Errors 1 and 2 had each, at the time, supported a conclusion that this framework's small-cohort setting could not learn under genuine privacy constraints; Error 3 subsequently invalidated the corrected headline number this conclusion's reversal had produced. Section VI reports only results obtained after all three corrections.

## VI. Experimental Results

### A. Setup

Results use 20 simulated UAVs, a Dirichlet($\alpha = 0.5$) non-IID partition of each dataset, batch size 32 (4 for AIDER), 5 local epochs, gradient clipping at $C = 1$, and a target $(\varepsilon_{\max} = 10, \delta = 10^{-5})$-DP guarantee. CIFAR-10 runs use 200 communication rounds; EuroSAT and

TABLE I
CIFAR-10, final accuracy, 200 rounds, 5 seeds.

| Seed | 42 | 43 | 44 | 45 | 46 |
|---|---|---|---|---|---|
| Uniform $\sigma$ | 17.3% | 21.9% | 18.7% | 18.4% | 10.0% |
| Adaptive $\sigma$ | 35.4% | 31.7% | 17.8% | 34.9% | 33.7% |

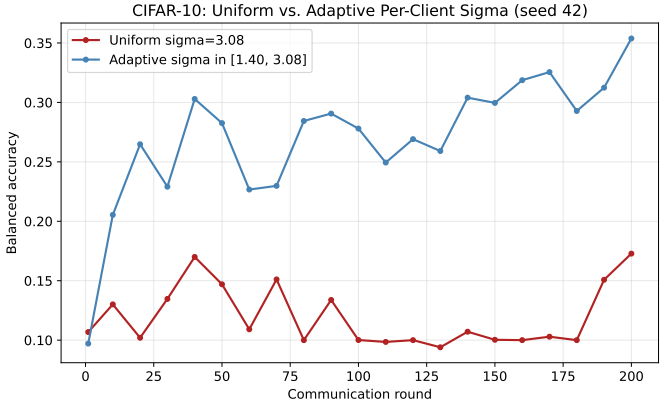



Fig. 1. CIFAR-10 balanced accuracy, uniform vs. adaptive per-client $\sigma$, seed 42, 200 rounds, both at an identical formal $\varepsilon \leq 10$ guarantee.

TABLE II
EuroSAT, final balanced accuracy, 30 rounds, 3 seeds.

| Seed | 42 | 43 | 44 |
|---|---|---|---|
| Uniform $\sigma$ | 38.1% | 48.5% | 28.1% |
| Adaptive $\sigma$ | 51.9% | 61.2% | 50.5% |

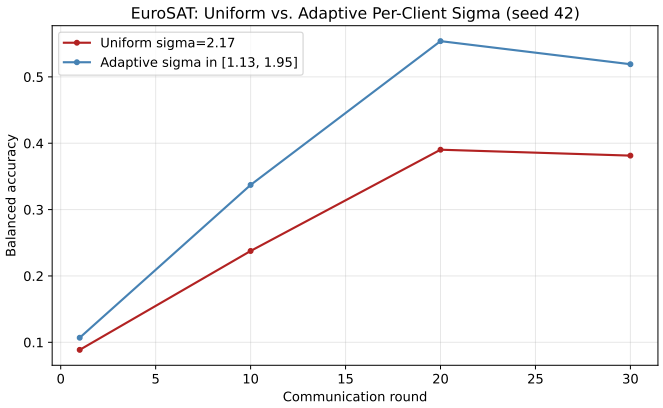



Fig. 2. EuroSAT balanced accuracy, uniform vs. adaptive per-client $\sigma$, seed 42, 30 rounds.

AIDER use 30. Each client's correct $\sigma$ (for the uniform-$\sigma$ baseline) or $\sigma_i$ range (for adaptive calibration) is re-derived per random seed against that seed's actual Dirichlet partition, since the worst-case client's size varies by draw. All training uses per-sample DP-SGD gradients computed via vectorised vmap, on a single NVIDIA GB10 GPU.

### B. CIFAR-10: Uniform vs. Adaptive $\sigma$

Table I reports final accuracy across 5 independent seeds. Uniform-$\sigma$ DP-FL achieves $17.3\% \pm 3.9$ mean final accuracy, frequently at the 10% chance level for individual checkpoints within a run. Adaptive calibration raises this to $30.7\% \pm 6.6$, winning 4 of 5 seeds by a substantial margin; the one exception (seed 44, $-0.9$pp) is an approximate tie, not a loss. We tested, and rejected, a hypothesis that this gain tracks a partition's client-size heterogeneity ratio: the ordering did not hold across all 5 seeds, and no single-variable predictor of the seed-to-seed variance was found. The honest characterisation is a substantial, mostly consistent mean improvement at an identical formal guarantee, not a deterministic function of one measured partition statistic.

### C. EuroSAT: Uniform vs. Adaptive $\sigma$

Table II reports the same comparison on EuroSAT (30 rounds, balanced accuracy). Adaptive calibration wins all 3 seeds tested, raising mean balanced accuracy from $38.2\% \pm 8.3$ to $54.5\% \pm 4.8$ (+16.3pp). That this effect size (+16.3pp) is close to CIFAR-10's (+13.4pp) despite the two datasets differing in resolution, class count, class balance, and round budget is evidence the effect generalises across datasets rather than being an artefact of one experiment.

### D. AIDER: A Negative Result That Narrows the Diagnosis

On AIDER, a real UAV disaster-imagery dataset with severe class imbalance (68% majority class) and a very small worst-case client (61 of the fleet's smallest partition's samples), adaptive calibration provides no benefit: balanced accuracy remains at the 20% chance level for 5 classes under both uniform (19.9% mean) and adaptive (19.7% mean) calibration. Because FedAvg already weights each client's contribution by its sample count, the tiny worst-case clients adaptive calibration targets barely influence the aggregate either way; this negative result rules out noise magnitude as AIDER's bottleneck and narrows the real cause to class imbalance interacting with the non-IID partition itself, motivating class-weighted loss rather than further noise-calibration work as the next step for this dataset.

### E. Comparison to Published DP-FL Results

Table III places this work's CIFAR-10 results against two recent client-level DP-FL papers. Adaptive calibration roughly halves the gap to these published numbers without changing the formal guarantee; the remaining gap is attributable to an order-of-magnitude smaller client pool (20 UAVs vs. 500 clients) and a lighter model (SimpleCNN vs. ResNet-18/GNResNet-10), not to the noise-allocation problem this paper addresses.

### F. Blockchain, Incentive, and MARL Results

A 40-round full-stack simulation (all layers active) shows the blockchain layer correctly tracking one block per round with no stalls, 2000 tokens distributed consistent with the Stackelberg equilibrium effort levels, and both the injected

TABLE III
Published client-level DP-FL results on CIFAR-10.

| Method | Setting | Acc. |
|---|---|---|
| DP-FedSAM [5] | 500 clients, non-IID | 53.9% |
| DP-FedPGN [6] | non-IID | 58.2% |
| This work, uniform $\sigma$ | 20 UAVs | 17.3%±3.9 |
| This work, adaptive $\sigma$ | 20 UAVs | 30.7%±6.6 |

Byzantine and free-rider attackers excluded from the top-8 token leaderboard within 1–2 rounds of participation. In the MARL layer, each UAV's participation policy, initially volatile during DQN exploration (15–90% participation), locks to 90–95% once the exploration floor is reached, with the cumulative privacy budget's growth rate tracking this participation variability directly, a property unavailable to static, non-learned participation scheduling.

## VII. Discussion and Limitations

The absolute accuracy levels reported here are modest relative to large-pool DP-FL literature; this is expected and, we argue, the right comparison point given the setting: a 20-client, channel-gated cohort with per-client independent accounting is intrinsically a harder noise regime than a 500-client pool, and the contribution of this paper is the noise-allocation improvement within that regime, not a claim of state-of-the-art absolute accuracy. All results are simulation-based on a single hardware platform; we have not yet validated the mechanism over an actual wireless UAV testbed, nor tested cohorts larger than 20 UAVs, both queued as future work. The adaptive-clipping mechanism of Andrew et al. [14], evaluated separately in the accompanying technical report, has not yet been tested in combination with adaptive $\sigma$; whether the two compose is an open question. Finally, the seed-to-seed variance in CIFAR-10's gain (+9.9 to +23.7pp) indicates the mechanism's benefit, while consistently positive, is itself a function of the realised client-size distribution in ways not yet fully characterised.

## VIII. Conclusion

A single, population-wide noise multiplier, calibrated to a fleet's smallest client, is the wrong way to add differential privacy to federated learning over a small, heterogeneous UAV cohort: at a genuinely valid per-client guarantee, it leaves most of this setting's achievable accuracy on the table. Per-client adaptive noise calibration, which exploits the already-independent structure of DP accounting to give each client only the noise its own data requires, recovers substantial learning at an identical formal guarantee, confirmed across two datasets and eight random seeds, while correctly providing no benefit where the real bottleneck is not noise at all. Integrated with a blockchain security layer, a game-theoretic incentive mechanism, and a MARL-driven participation policy, this gives a complete, privacy-rigorous data-sharing stack for UAV emergency-response networks, built on a correction history we report in full because the failure mode behind it, a plausible but wrong privacy accountant, is easy to repeat.

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